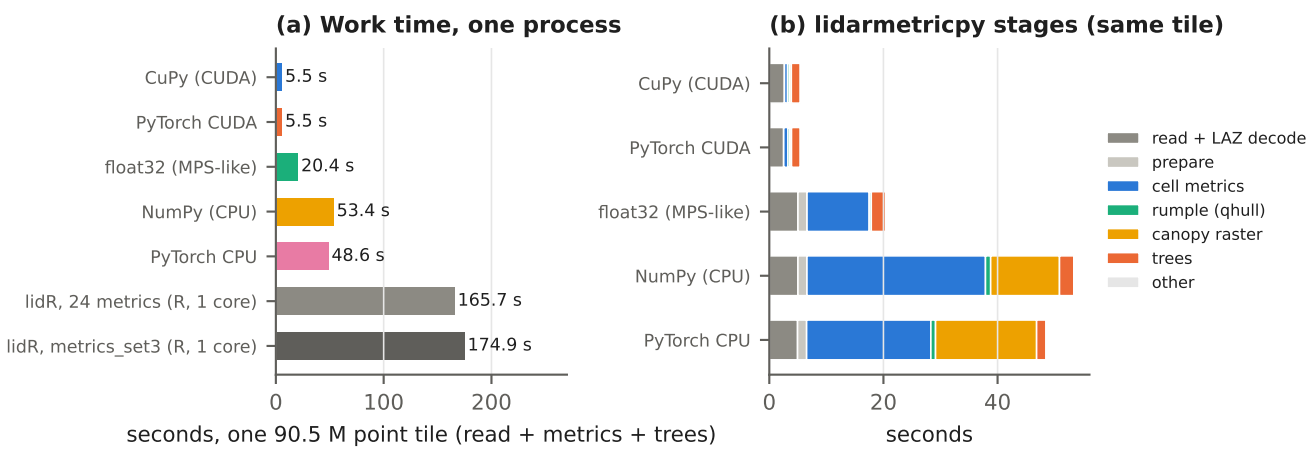
(a) Work time, one process
CuPy (CUDA)
PyTorch CUDA
float32 (MPS-like)
NumPy (CPU)
PyTorch CPU
lidR, 24 metrics (R, 1 core)
lidR, metrics_set3 (R, 1 core)
5.5 s
5.5 s
20.4 s
53.4 s
48.6 s
165.7 s
174.9 s
0
100
200
seconds, one 90.5 M point tile (read + metrics + trees)
(b) lidarmetricpy stages (same tile)
CuPy (CUDA)
PyTorch CUDA
float32 (MPS-like)
NumPy (CPU)
PyTorch CPU
0
20
40
seconds
read + LAZ decode
prepare
cell metrics
rumple (qhull)
canopy raster
trees
other